

Flag Feasibility Simulator — Model Assumptions

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This document summarizes every assumption built into the Flag Feasibility Simulator: where the underlying actuals came from, how the model turns those actuals into projected outcomes, and the specific default numbers currently in use for Hotel 1620 and Inn at Middletown.

1. Data Sources (per property)

Hotel 1620 (Plymouth Harbor, Plymouth MA — 177 rooms)

- Occupancy, ADR, and comp-set benchmarks: STR “Tab 4 — Competitive Set Report,” Running 12 Month column (trailing to May 2026) — the most representative full-year figure available, rather than a single month's snapshot.
- Group vs. transient occupancy split: derived from the BOB (Business on the Books) report, Running 12 Month (Jun 2025–May 2026), using the actual room-night mix (group rooms sold vs. individual rooms sold as a share of total rooms sold). That proportion is applied to the STR-sourced total occupancy figure, since STR and BOB's own totals didn't match exactly (STR is treated as the more reliable total).

Inn at Middletown (Middletown, CT — 100 rooms)

- Occupancy, ADR, and comp-set benchmarks: same STR “Comp” tab, Running 12 Month methodology as Hotel 1620.
- Group vs. transient occupancy split: derived from the ROB Master Workbook's revenue split (group vs. transient revenue), averaged across its two available trailing-year columns, converted to a room-night proportion and applied to the STR total occupancy figure the same way as 1620.

Bottom line: for both properties, STR determines “how big is the pie” (total occupancy, ADR, comp set), and the property-specific revenue system (BOB for 1620, ROB for Middletown) determines “how the pie splits” between group and transient business.

2. Current Actuals Baked Into the Model

	Hotel 1620	Inn at Middletown
Rooms	177	100
Occupancy	49.86%	47.18%
Comp set occupancy	57.15%	57.74%
ADR	\$186.50	\$219.44
Comp set ADR	\$172.88	\$154.64
Transient occupancy	22.02%	25.67%
Group occupancy	27.72%	21.51%
Gap to comp set	7.29 pp	10.56 pp

(Every field above is editable in the app — these are the current defaults, not hardcoded ceilings.)

3. How the Model Projects Each Flag Scenario

For each flag type (Hard: Marriott/Hilton/IHG-tier; Soft: Autograph/Tapestry/Vignette-tier), the model takes five base-case assumptions you set directly:

- Franchise + royalty fee (% of gross revenue)
- Expected transient occupancy lift (percentage points)
- Expected group occupancy lift (percentage points)
- ADR impact (%)
- PIP (property improvement plan) cost per room

From that single base case, “Optimistic” and “Underperforming” variants are generated automatically (not independently estimated):

- Transient/group lift swing $\pm 50\%$ of their own value (so hard flag's +12pp base becomes +18pp optimistic / +6pp underperforming)
- Fee multiplies $\times 0.85$ (optimistic) / $\times 1.15$ (pessimistic)
- PIP cost multiplies $\times 0.70$ (optimistic) / $\times 1.30$ (pessimistic)
- ADR impact swings by a flat ± 2.5 percentage points

Then revenue = (base transient occ + lift) $\times$ room-nights $\times$ ADR, same for group, minus fee% of gross, minus PIP:

- Room-nights available = rooms $\times$ 365
- Transient revenue = (current transient occ% + transient lift) $\times$ room-nights $\times$ scenario ADR
- Group revenue = (current group occ% + group lift) $\times$ room-nights $\times$ scenario ADR
- Gross revenue = transient + group revenue
- Fee cost = gross revenue $\times$ franchise fee %
- PIP cost = (PIP \$/room $\times$ rooms) amortized straight-line over the chosen number of years (7 years by default)
- Net revenue = gross revenue – fee cost – amortized PIP cost
- Compared against “Stay Independent” (current actuals, zero fee, zero PIP) for net revenue vs. baseline and a payback period in years.

Current default flag assumptions (base case, fully editable per property):

	Fee	Transient lift	Group lift	ADR impact	PIP cost/room
Hard flag	11.0%	+12.0pp	+0.0pp	–3.0%	\$18,000
Soft brand	7.0%	+6.0pp	+0.0pp	+1.0%	\$7,500

PIP amortization period: 7 years (default, adjustable 1–20 years)

4. Key Structural Assumptions / Limitations

- Group occupancy lift defaults to 0pp for both flag types — the model assumes group/meetings business is relationship- and market-driven, not brand-driven, and is held flat unless you explicitly set a group lift.
- Transient and group segments are assumed to earn the **same ADR** in the revenue math — there's no separate transient-rate vs. group-rate assumption. (This means how you split an occupancy gain between transient/group doesn't change total revenue, only which line item it shows up under.)
- There is currently no linkage between PIP dollars spent and how much occupancy or ADR actually improves — the occupancy lift and PIP cost are set independently, not tied to each other by a formula.
- The occupancy-lift assumptions (the +12pp/+6pp defaults above) are manually set, not derived from the comp-set occupancy gap. An alternative approach — tying expected occupancy lift to a defined

percentage of the actual gap to comp set — was discussed and tested outside the live app, but is not yet implemented in the tool.

- Comp-set figures (occupancy/ADR) are currently informational/display only in the app — they are not yet wired into the revenue projections.

5. What's Editable vs. Fixed

Editable per property, live in the app: room count, current occupancy/ADR/comp set, transient/group split, all five base-case flag assumptions per flag type, PIP amortization period.

Fixed by the model's internal formulas (not user-facing sliders): the $\pm 50\%$ optimistic/underperforming swing rule, the fee/PIP multiplier ranges, and the ADR swing of $\pm 2.5\text{pp}$ described in Section 3.